FUNDEV INTERNATIONAL COMPETITION 2026 — USU MEDAN | GEODESIC CADASTRE & MASS-BALANCE ENGINE | REGULATION (EU) 2023/1115 ART. 9 | CASE STUDY: WEST LAMPUNG & TNBBS UNESCO CORRIDOR

TERRAPROOF: Offline-First Geodesic Polygon & Biological Mass-Balance Compliance Engine

Eliminating Tropical Canopy GPS Multipath Distortion & Intercepting Illicit Rainforest Harvest Laundering for 2.5M Indonesian Smallholder Exporters

SDG 2.3 & 2.4 +22% Smallholder Income	**SDG 1.4** Cadastral Export Lock
SDG 17.11 EU Traces-NT Bridge	**SDG 15.2** 0.00% TNBBS Intrusion

FAILURE MODE 01 — TROPICAL CANOPY L1 GPS MULTIPATH — FIELD TELEMETRY ANALYSIS

False Rejection of Honest Smallholders (30.67% Area Error)

Under dense shade trees (*Gliricidia* / *Erythrina*) and steep volcanic valleys in West Lampung, consumer smartphone L1 GPS signals bounce off wet foliage—jumping **15m to 55m** in 1 second. Raw tracks create self-intersecting polygons that falsely overlap protected forest borders, threatening **$2.1B+ in Indonesian smallholder exports**.

UNFILTERED FIELD TRACK	TERRAPROOF SANITIZED
30.67% Area Distortion	0.00% Error (100% Recall)

FAILURE MODE 02 — THE "POLYGON LAUNDERING" LOOPHOLE — SUPPLY-CHAIN AUDIT

Illicit National Park Coffee Smuggled via Compliant Polygons

Existing EUDR tools (FAO Open Foris Whisp, EU Traces-NT) only validate static 2D map coordinates. Middlemen exploit this by registering a tiny **0.82 ha compliant boundary plot** and funneling **2,500+ kg of illegal coffee** harvested inside **Bukit Barisan Selatan National Park (TNBBS)** through that single clean polygon ID.

STATIC GIS CHECK ONLY	Y_{max} MASS-BALANCE LOCK
0% Fraud Interception	100% Fraud Blocked

SYSTEM ARCHITECTURE — THREE DETERMINISTIC ALGORITHMIC ENGINES (DUAL PHP + MYSQL 8 SPATIAL) — ZERO EXTERNAL PAID API DEPENDENCY

ENGINE 01 • SPATIAL & KINEMATIC — 0.57 ms / track

Canopy Multipath & Geodesic Clipper

Strips L1 carrier reflection spikes via kinematic velocity (v_i) and interior deflection angle (θ_i), computes WGS84 Spherical Excess area, and clips against TNBBS via Sutherland-Hodgman:

$$\text{Spike iff: } v_i > 3.5 \text{ m/s} \vee (\theta_i < 18^\circ \wedge d_i > 12\text{m})$$
$$A_{ha} = (R^2/2) \left|\Sigma \text{ rad}(\Delta\lambda) \left[\sin(\varphi_2)+\sin(\varphi_1)\right]\right| \times 10^{-4}$$

ENGINE 02 • AGRONOMIC MASS-BALANCE — 100% Precision

Biological Yield-Ceiling (Y_{max}) Lock

Computes the maximum biologically plausible dry harvest per plot from geodesic area, tree density, physiological age curve $\phi_{age}(t)$, elevation $\psi_{elev}(h)$, and intercrop factor $\kappa_{intercrop}$:

$$Y_{max} = A_{ha} \times D_{trees} \times y_{base}(c) \times \varphi_{age}(t) \times \psi_{elev}(h) \times \kappa_{intercrop}$$
$$\text{If } W_{kg} > Q_{rem} \Rightarrow \text{BLOCKED_LAUNDERING (Quota Preserved)}$$

ENGINE 03 • CRYPTOGRAPHIC COMPILER — RFC 7946 CRS84

SHA-256 Merkle & Traces-NT Compiler

Chains every field weigh-in into a tamper-evident SHA-256 ledger ($0 gas fees), builds a binary Merkle Root per container, and exports 6-decimal GeoJSON for EU customs:

$$H_k = \text{SHA256}(H_{k-1} \,||\, \text{plot} \,||\, W_{kg} \,||\, \text{verdict} \,||\, \text{ts})$$
$$\text{MerkleRoot} = \text{SHA256}(H_{2i} \,||\, H_{2i+1}) \Rightarrow \text{Traces-NT DDS}$$

END-TO-END PIPELINE & EMPIRICAL DATASET — KOPERASI PRODUSEN KOPI LESTARI LAMPUNG — 15 PARCELS • 12 FARMERS • 25 SHA-256 BLOCKS • 2 EU CONTAINERS

01 / OFFLINE CANOPY WALK
Collector walks farm perimeter in blank-spot rainforest. GPS track & scale weigh-ins queue in browser **IndexedDB / localStorage**.

02 / GEODESIC CLIPPING
Strips 56 multipath spikes across 15 plots; locks **3 TNBBS deforestation plots** (6.8%–24.5% clip) & **2 neighbor overlaps**.

03 / ANTI-LAUNDERING LOCK
Intercepts **5 quota laundering attacks** (7,630 kg blocked, e.g., 1,650 kg injected into LP-PLT-001 with 340 kg quota left).

04 / TRACES-NT PASSPORT
Compiles 20 approved harvests (8,463 kg) into **2 Merkle-verified containers** for Hamburg (DE) & Trieste (IT).

EMPIRICAL VALIDATION (`PHP ARTISAN TERRAPROOF:BENCHMARK`) — 14/14 PHPUNIT TESTS PASSING (74 ASSERTIONS)

BENCHMARK SUITE	DATASET	VERIFIED RESULT	LATENCY
1. GPS Spike Sanitizer	50 Tracks (199 Spikes)	30.67% → 0.00% Err	0.57 ms
2. Sutherland-Hodgman Clip	50 Polygons vs TNBBS	100.0% Accuracy	3.44 ms
3. Y_{max} Anti-Laundering	100 Weigh-Ins (30 Fraud)	100% Prec / 100% Rec	0.0018 ms
4. RFC 7946 & SHA-256 Chain	25 Blocks / 2 Containers	100% Intact & Valid	<1.20 ms

COMPETITIVE MATRIX VS. GLOBAL BASELINES — EUDR COMPLIANCE SYSTEMS

CAPABILITY	EU TRACES-NT	FAO WHISP	TERRAPROOF
Canopy Multipath Spike Filter	No (Rejects)	No	YES (100% Recall)
Biological Yield (Y_{max}) Lock	No	No	YES (Mass-Balance)
Offline Blank-Spot Queue	Cloud Only	Cloud Only	YES (IndexedDB)
Cost per Smallholder Farmer	Portal Only	API Only	$0.00 (Co-op SaaS)

SDG 2.3 SMALLHOLDER PREMIUM
+18% to +24% EU Export Price

SDG 15.2 RAINFOREST PROTECTED
TNBBS & Register 39 Corridor

COOPERATIVE DEPLOYMENT COST
Rp 2.5M/yr ($155/yr) • 92% Savings

FunDev<> International Competition 2026 • Faculty of Vocational Studies, Universitas Sumatera Utara
Deterministic Production Build • Laravel 13 + WGS84 Spherical Excess + SHA-256 Merkle Engine